

Soldiers Surf Free

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Background & Motivation

Ido Aviv was head of surf coaching at the Kiryat Yam surf club. He used to hand a surfboard to any soldier who walked up — *"take a surfboard, I trust you."*

After Ido fell in the war, his family and the Kiryat Yam Surf Club set out to scale that gesture: let any IDF soldier borrow a surfboard, unattended, at every participating beach.

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The Problem

- ◆ **Unstaffed beaches.**
- ◆ **Soldier verification with no staff.**
- ◆ **Many sites, lockers and cells.**
- ◆ **Weatherproof, tamper-resistant hardware.**

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Our Solution — an end-to-end lending platform

Soldier

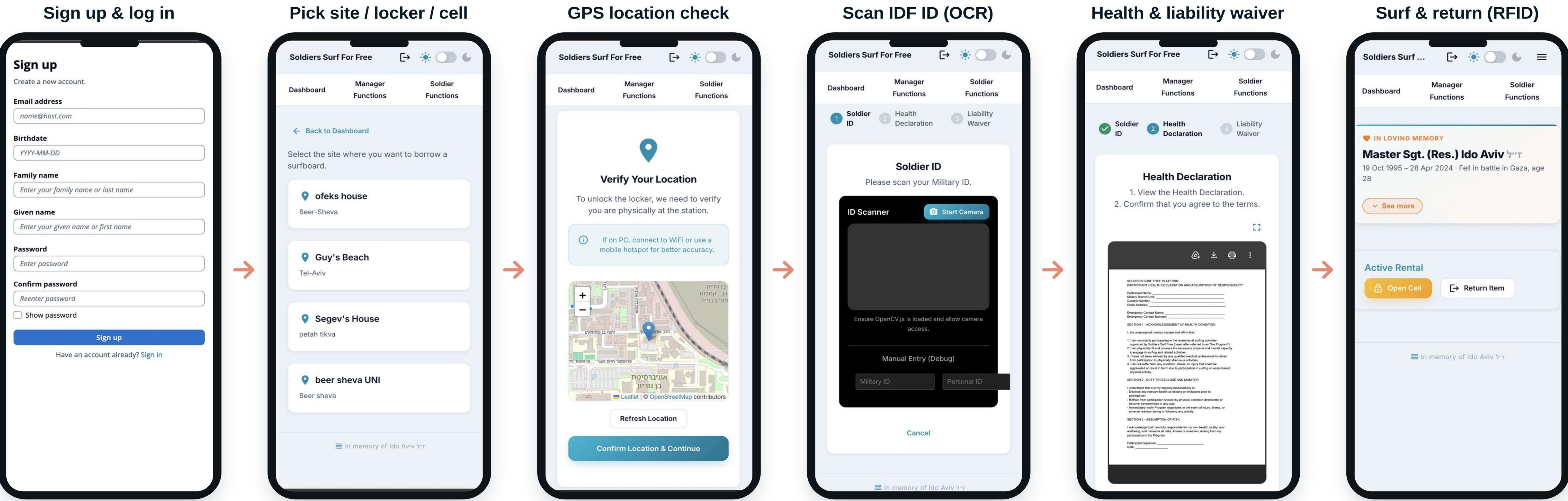
Site Manager

Admin

Soldiers Surf Free is a web platform that runs unattended surfboard lending across many beaches. Soldiers borrow and return boards, site managers oversee their site and control lockers remotely, and admins manage users, inventory and sites. A FastAPI backend enforces each site's policy, while smart ESP32 lockers talk to it over MQTT through AWS IoT. Identities are verified once and stored encrypted, so no staff is ever needed at the beach.

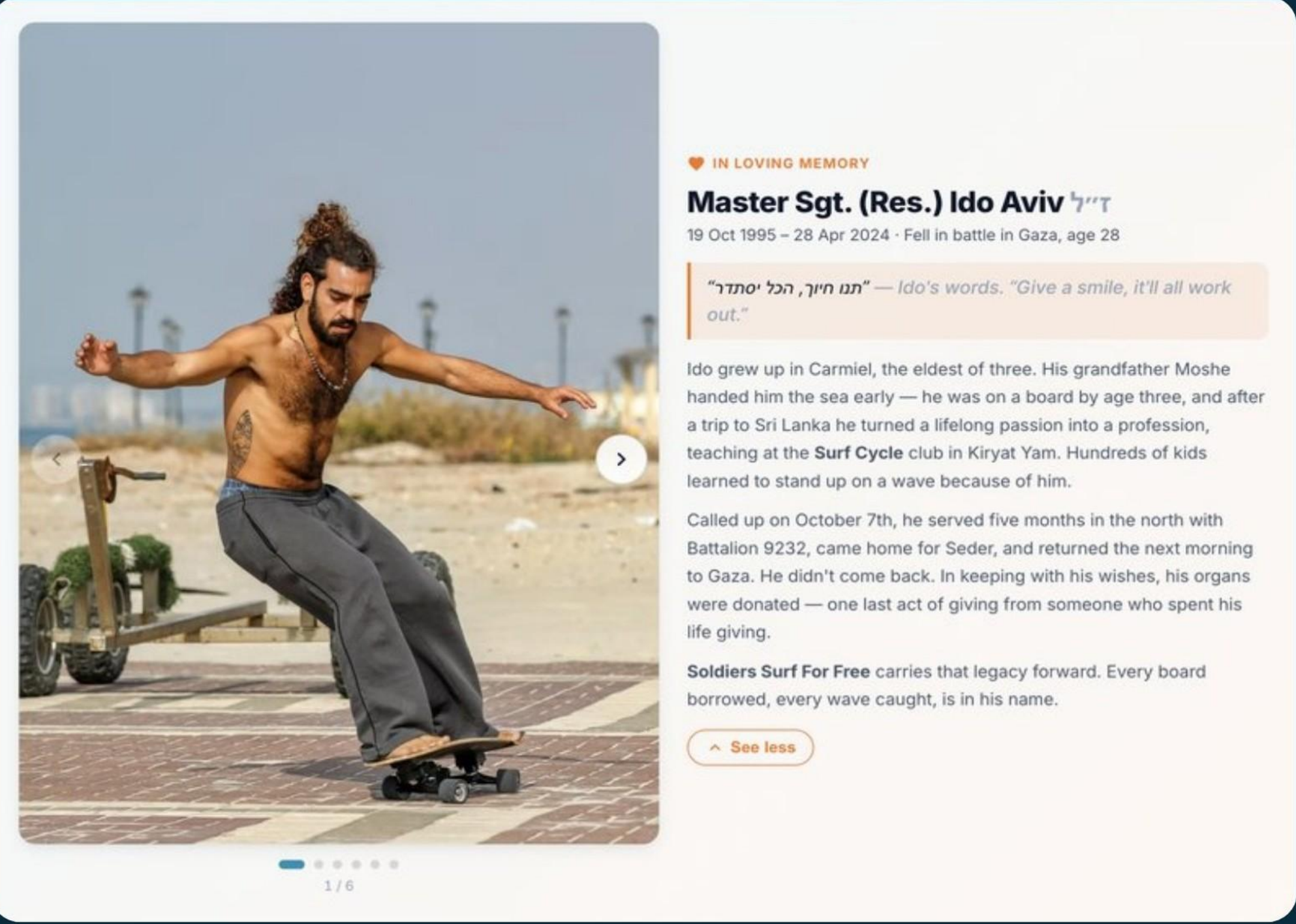
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How a borrow works



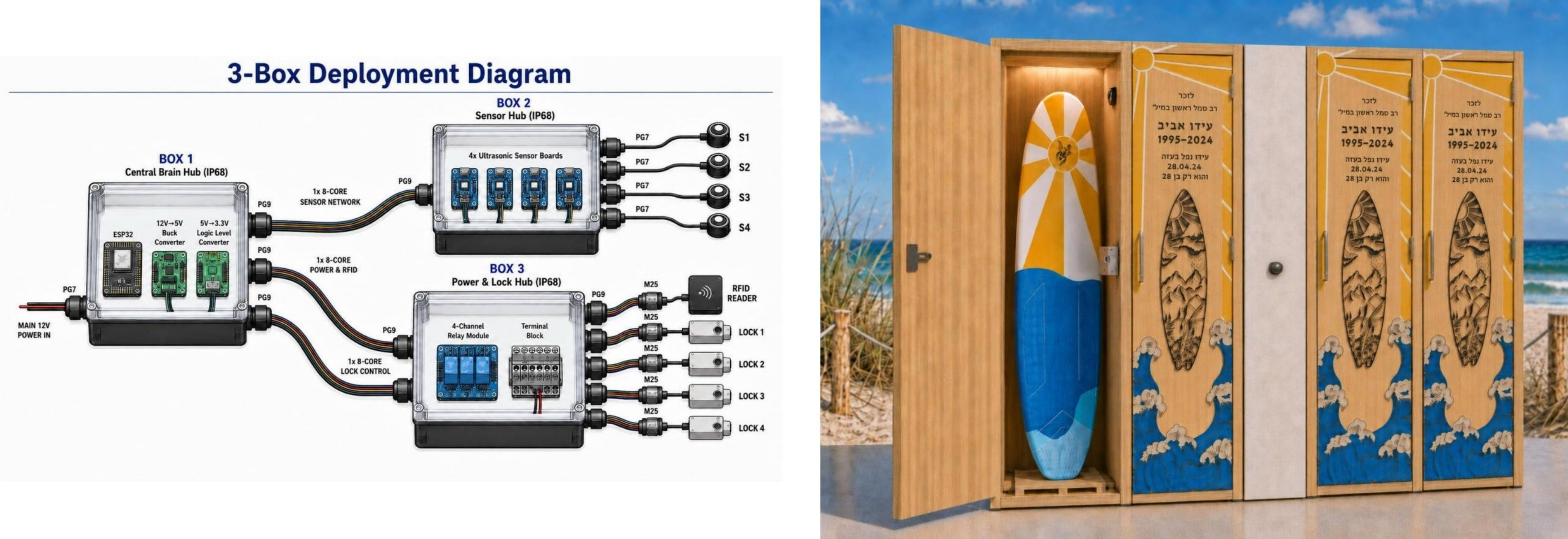
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Meet Ido



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Hardware



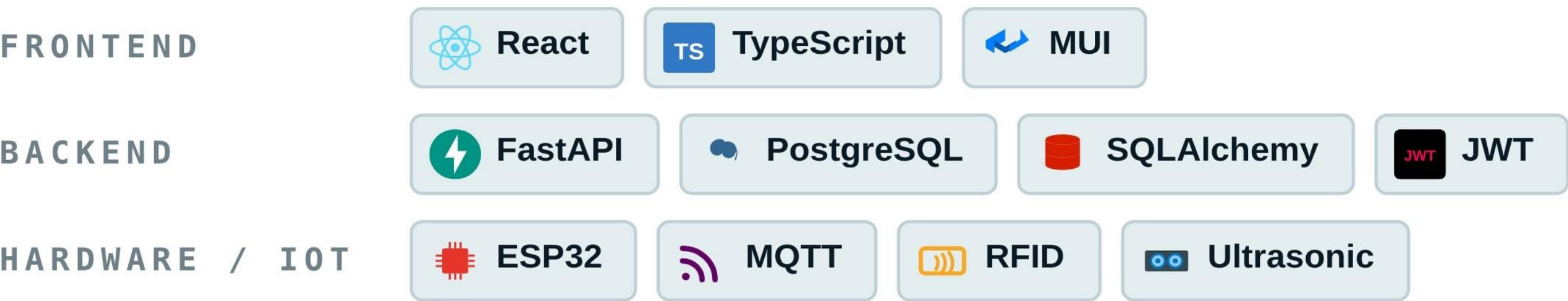
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Methodology

- ◆ **Requirements built with the customer**
- ◆ **Per-site policy enforced server-side**
- ◆ **Browser OCR ID verification**
- ◆ **Real ESP32 locker testing**

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Tech Stack



9

Conclusions & Next Steps

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End-to-end works
Borrow and return run end-to-end across soldier, manager and admin roles on a real ESP32 cabinet.
- 🌐

Scales to many beaches
Per-site policy enforced server-side — adding a beach is configuration, not a new build.
- +

Next: scale the network
More cabinets, richer site analytics, a donation flow, and offline-tolerant lock firmware.